DocChat Test Document

Artificial Intelligence, Large Language Models, and Retrieval-Augmented Generation

A practical sample PDF for testing document upload, text extraction, chunking, retrieval, and question answering in a DocChat application.

Document type: Educational reference guide
Suggested use: Upload this PDF to your DocChat app and ask questions whose answers are contained in the document.
Length: 12 pages (within the 30-page limit).

This document is a synthetic learning resource created for application testing. It is not a formal standard or a substitute for official technical documentation.

1. Introduction to Artificial Intelligence

Learning objective: Understand the key ideas and apply them when building or evaluating a document question-answering application.

What is AI?

Artificial Intelligence (AI) is a field of computing focused on building systems that perform tasks associated with human intelligence. These tasks can include recognizing patterns, understanding language, making predictions, planning, and solving problems. AI systems vary widely: a rule-based program follows explicit instructions, while a machine-learning model learns statistical patterns from examples.

Common AI categories

Narrow AI is designed for a particular task, such as spam detection, translation, or product recommendations. General AI is a hypothetical capability to perform a broad range of intellectual tasks at a human-like level. Most deployed AI systems today are narrow systems, even when they can support many related workflows.

AI in everyday applications

Examples include email spam filters, voice assistants, navigation estimates, fraud alerts, image search, customer-support chatbots, and predictive maintenance. The quality of an AI system depends on its data, design, evaluation, and the context in which it is used.

Important limitation

An AI output can be incorrect, incomplete, biased, or out of date. A reliable application should communicate uncertainty, protect private data, test performance, and provide a way for users to verify important claims.

2. Machine Learning Fundamentals

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Definition

Machine Learning (ML) is a branch of AI in which algorithms learn patterns from data to make predictions or decisions without requiring every rule to be manually programmed. Training adjusts a model's parameters to reduce a chosen error or loss measure.

Supervised learning

Supervised learning uses examples containing inputs and known target outputs. Classification predicts a category, such as whether a transaction is fraudulent. Regression predicts a numeric value, such as tomorrow's energy demand.

Unsupervised learning

Unsupervised learning looks for structure in data without target labels. Clustering can group similar customers, while dimensionality reduction can represent complex data using fewer variables.

Reinforcement learning

Reinforcement learning trains an agent to choose actions through interaction with an environment. Feedback is expressed as rewards or penalties, and the objective is to learn a policy that maximizes cumulative reward.

Training, validation, and test data

Training data is used to fit the model. Validation data helps choose settings and compare model versions. Test data estimates performance on data that was not used for those choices. If the test set influences repeated tuning, it no longer provides an unbiased final evaluation.

Overfitting

Overfitting occurs when a model learns details or noise specific to its training examples and performs poorly on new data. Common responses include collecting more representative data, simplifying the model, regularization, and cross-validation.

3. Neural Networks and Deep Learning

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Neural networks

A neural network contains connected computational units arranged in layers. Each unit combines inputs using weights and a bias, applies an activation function, and passes the result forward. During training, backpropagation calculates gradients and an optimizer updates parameters.

Deep learning

Deep learning uses neural networks with multiple layers to learn complex representations. It is useful for images, audio, text, and other high-dimensional data, but can require substantial data, compute, and careful evaluation.

Common architectures

Convolutional neural networks are often used for image-like data. Recurrent neural networks process sequences step by step. Transformers use attention mechanisms to model relationships among tokens and are widely used in modern language models.

Training vocabulary

An epoch is one complete pass through the training dataset. A batch is a subset of examples processed together. The learning rate controls the size of optimizer updates. A loss function measures the mismatch between predictions and targets.

4. Natural Language Processing

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Definition

Natural Language Processing (NLP) concerns methods that allow computers to process human language. Tasks include classification, sentiment analysis, translation, summarization, named-entity recognition, information extraction, and question answering.

Tokenization

Tokenization splits text into units called tokens. A token may be a word, part of a word, punctuation, or another text unit. Language models commonly operate on token IDs rather than raw characters.

Embeddings

An embedding is a numerical vector representing an item such as a word, sentence, paragraph, or document. Similarity between embeddings can help retrieve text with related meaning, including when the wording differs.

Evaluation

NLP evaluation depends on the task. Classification may use precision, recall, and F1 score. Summaries may require human review as well as automated measures. A fluent answer is not necessarily a factually correct answer.

5. Large Language Models

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What is an LLM?

A Large Language Model (LLM) is a neural network trained on large collections of text and sometimes other modalities. Many LLMs learn to predict the next token and can then generate text, answer questions, summarize information, explain code, or follow instructions.

Prompt and context

A prompt is the instruction or input provided to a model. The context window is the amount of tokenized information the model can consider in one request. If important information is absent from the prompt or context, the model may be unable to use it reliably.

Temperature

Temperature affects the randomness of token selection during generation. Lower values generally make outputs more predictable; higher values can produce more varied responses. Temperature does not guarantee correctness.

Hallucination

A hallucination is an output that sounds plausible but is unsupported or incorrect. To reduce this risk, applications can provide trusted source material, require citations, validate structured outputs, and instruct the model to say when the answer is not found.

Fine-tuning versus prompting

Prompting changes the instructions supplied at inference time. Fine-tuning updates model parameters using additional examples. Fine-tuning can help shape behavior or task performance, while it is not automatically the best way to add frequently changing factual knowledge.

6. Retrieval-Augmented Generation (RAG)

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Definition

Retrieval-Augmented Generation combines information retrieval with text generation. Instead of relying only on the model's learned parameters, a RAG application searches a collection of documents and places relevant passages into the prompt used to generate an answer.

Typical pipeline

A common pipeline ingests documents, extracts text, splits text into chunks, creates searchable representations, stores chunks and metadata, retrieves relevant chunks for a user query, constructs a prompt with the retrieved context, and generates an answer.

Chunking

Chunking divides long documents into smaller passages. Very large chunks may contain irrelevant material; very small chunks may lose context. Chunk size and overlap should be tested on the application's documents and questions.

Embeddings and vector search

An embedding model maps text to vectors. A vector database or index can find passages whose vectors are close to the query vector under a similarity measure. Metadata filters can restrict retrieval by source, date, topic, or access permissions.

Keyword and hybrid retrieval

Keyword retrieval matches terms and is useful for exact names, identifiers, and phrases. Vector retrieval can capture semantic similarity. Hybrid retrieval combines both signals and may improve recall across different query types.

Reranking

A reranker evaluates a set of retrieved candidates and reorders them for relevance to the query. It can improve the quality of the final context, though it adds latency and computation.

Grounded generation

A grounded answer should use the retrieved context rather than invent missing facts. The prompt should tell the model to answer from the supplied passages, distinguish evidence from inference, and state when the documents do not contain the answer.

7. Vectorless and Document-Based Retrieval

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What does vectorless mean?

In a vectorless retrieval design, the application does not use vector embeddings as its primary retrieval mechanism. It may use keyword search, full-text search, metadata filters, structured queries, or direct extraction from a small document. The term is used informally and can refer to different architectures.

When it works well

Keyword or full-text search can work well when users know relevant terms, document collections are modest, and exact phrases, identifiers, or named entities matter. A small PDF may sometimes be searched directly without building a vector index.

Trade-offs

Vectorless methods can be simpler and cheaper to operate, but may miss relevant passages when the query uses different wording from the document. Vector retrieval can help with semantic matches but introduces embedding and index-maintenance requirements.

A practical strategy

Start with the simplest retrieval approach that meets evaluation needs. Create a set of representative questions, record whether the correct passage is retrieved, and compare keyword, vector, or hybrid methods. Choose based on measured answer quality, latency, and cost rather than the method's label.

8. Building a DocChat Application

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Document ingestion

A DocChat application accepts a document, extracts its text, stores useful metadata, and makes the content searchable. Common inputs include PDF, TXT, DOCX, and Markdown, although supported formats depend on the implementation.

PDF text extraction

Text-based PDFs often allow direct extraction. Scanned PDFs may contain only page images and require OCR. Tables, columns, headers, footers, and unusual fonts can cause extraction errors. Inspect extracted text before blaming retrieval or generation.

Metadata

Useful metadata can include filename, page number, section title, upload timestamp, document ID, and access-control scope. Metadata enables filtering and helps the application cite the original source.

Question-answering flow

The user submits a question. The app retrieves likely relevant passages, combines them with instructions, sends the request to an LLM, and displays the answer. A useful interface can also show the source passages or page numbers.

Conversation memory

Conversation history can help interpret follow-up questions, but it should not replace document retrieval. The system should distinguish the user's current question, relevant conversation history, and evidence retrieved from the document.

Security

Treat uploaded files and their text as untrusted input. Avoid executing document content as code. Enforce per-user document permissions, validate file types and sizes, handle secrets securely, and consider prompt injection attacks embedded in documents.

9. Evaluating a RAG or DocChat System

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Retrieval quality

Measure whether the retrieved passages contain the evidence needed to answer a question. Recall at k asks whether relevant evidence appears among the top k results. Precision at k measures how many of those results are relevant.

Answer quality

Evaluate factual correctness, completeness, relevance, clarity, and whether claims are supported by the supplied context. Human review is important for ambiguous or high-impact tasks.

Citation quality

A citation should point to the passage or page that supports the statement. A citation that merely points to a document without supporting the claim can mislead users.

Test set

Build a test set containing direct questions, paraphrased questions, multi-part questions, questions requiring two sections, and questions whose answers are absent. Include questions about names, numbers, dates, and definitions.

Failure analysis

If an answer is wrong, check the stages in order: Was the text extracted correctly? Was the right chunk created? Did retrieval return it? Was it truncated from the prompt? Did the model ignore or misinterpret it? This staged diagnosis makes debugging more efficient.

10. Privacy, Ethics, and Responsible Use

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Privacy

Only upload information you are permitted to process. Minimize stored personal data, define retention rules, restrict access, and understand the data-handling terms of any external model provider.

Bias and fairness

Training data and design choices can produce uneven performance across groups or contexts. Test the system on representative examples and provide a way to report harmful or inaccurate outcomes.

Transparency

Explain what sources the system uses, whether answers are generated, and what the limitations are. When possible, show evidence and allow users to open the source document.

Human oversight

For legal, medical, financial, employment, or safety-critical decisions, generated answers should not be treated as the sole authority. Qualified human review and authoritative sources may be required.

11. Mini Case Study: A College Policy Assistant

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Scenario

A college wants students to ask questions about attendance rules, exam procedures, library hours, and scholarship deadlines. The source documents are PDFs published by authorized departments.

Design

The app extracts text from each PDF, preserves page numbers and document titles, chunks content by section, and indexes the chunks. Each query retrieves relevant passages and provides them to an LLM with an instruction to answer only from the evidence.

Example question

Question: How can a student find the exam registration deadline? A good system retrieves the exam notice or academic calendar, answers with the date if it is present, and cites the relevant document and page.

Missing information

If no uploaded document states the deadline, the system should say that it could not find the date in the available documents. It should not guess a date based on a previous semester.

Success criteria

The assistant should retrieve the correct policy, provide supported answers, cite the right pages, protect restricted documents, and clearly handle unanswered questions.

12. Glossary and Practice Questions

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Glossary

AI: systems performing tasks associated with intelligence. ML: learning patterns from data. NLP: computational processing of language. LLM: a large neural language model. Token: a unit processed by a model. Embedding: a vector representation. Chunk: a segment of a document. Retriever: a component that finds relevant evidence. Reranker: a component that reorders candidates. RAG: retrieval combined with generation. OCR: optical character recognition. Hallucination: unsupported or incorrect generated content.

Practice questions

1. What is the difference between AI and machine learning? 2. Why are training, validation, and test sets separated? 3. What is an embedding? 4. Explain the stages of a RAG pipeline. 5. Why does chunk overlap sometimes help? 6. How does keyword retrieval differ from vector retrieval? 7. What is a hallucination? 8. How can a DocChat system cite a PDF page? 9. What should the system do if the answer is absent? 10. Why is access control important for uploaded documents?

Suggested answers

1. ML is a subfield of AI that learns patterns from data. 2. Separation helps estimate generalization and avoid tuning to the test set. 3. An embedding is a numerical vector representing content. 4. Ingest, extract, chunk, index, retrieve, prompt, generate, and present evidence. 5. Overlap can preserve context across chunk boundaries. 6. Keyword retrieval matches terms; vector retrieval compares semantic representations. 7. It is plausible-sounding but unsupported or wrong output. 8. Store page metadata and show the page with each passage. 9. Say the available documents do not contain the answer. 10. It prevents unauthorized users from seeing private content.